Aashish Sunar

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Summary

Software engineer with three and a half years of shipping production web and backend code in shared repositories, with code review, automated tests, and CI. Used to owning a feature from the database schema through deployment and the bug report afterward.

Already build on the ArcGIS stack by choice: a natural language explorer over ArcGIS Feature Services using Esri web components and Calcite, a routing prototype on ESRI GeoAI that placed fourth at a 30-team hackathon, and a disaster alerting service running on NASA's live feed. The projects Esri lists for interns, test automation with language models, DevOps in Python and Docker, and disaster response apps on ArcGIS, are close to work I have already done.

Technical Skills

Languages: Python (proficient), Java (proficient), JavaScript (ES6+), TypeScript, C, C++, SQL, HTML5/CSS
Frameworks & Libraries: React, Next.js, Node.js, Express, FastAPI, REST API Design, pandas, NumPy
Geospatial: ArcGIS REST Services, ArcGIS web components, Calcite Design System, ESRI GeoAI, proximity and distance analysis, interactive web mapping
Testing & Quality: Playwright, headless browser automation, unit and integration testing, code review, shared repositories and branching workflows
DevOps & Tools: Git, GitHub, GitHub Actions, Docker, Linux, CI/CD, AWS (Lambda, API Gateway, S3, EC2), Google Cloud Platform, Microsoft Azure
Data & AI: PostgreSQL, MySQL, SQLite, MongoDB, Anthropic and OpenAI APIs, retrieval-augmented generation, LLM-as-judge evaluation, structured extraction from documents
Foundations: Data Structures, Algorithms, Object-Oriented Design, System Design, Async Processing, Agile and Scrum

Professional Experience

AI for Humanity Lab, Claremont Graduate University — Claremont, CA
Software Engineer | Research Fellow — *September 2025 to Present*

- Built a browser-based testing tool in TypeScript and Playwright that loads a live page and checks whether the AI disclosure notice required under EU AI Act Article 50 is actually visible to a person. The scanners already on the market look for that text in the page source and stop there, which passes an 8px gray notice sitting below the fold.
- Turned one line of legal wording, clear and distinguishable at the time of first interaction, into five things a machine can measure: where the notice sits on screen, its contrast against the background, its font size, whether anything covers it, and when it appears.
- The contrast checker returns an explicit unknown when the background is a gradient or an image and the true ratio cannot be resolved, because a test that guesses will eventually pass something it never measured.

Claremont Graduate University — Claremont, CA
Software Developer | Graduate Assistant — *August 2025 to Present*

- Built the academic advising chatbot in Python and FastAPI behind a REST API, working from the questions advisors were answering by hand, so a student gets a straight answer on program policy and degree requirements without waiting for office hours. Over 500 questions handled, and roughly 40 percent fewer of them reach a human advisor.
- Every answer passes a second automated check that reads it back against the official policy documents and holds anything those documents do not say. A student who acts on a wrong answer about degree requirements pays in time and tuition, which is worse than a slow reply.
- Wrote a classifier that sorts each question by what it is asking about and sends it to a prompt written for that kind of question. Answers were inconsistent before the routing, and that is what fixed it.
- The stages of the pipeline run at the same time instead of one after another, so an answer comes back in under two seconds, and a step that fails is retried rather than dropping the question.

Soka University of America Aliso Viejo, CA

Full Stack Developer | Office of Alumni Relations Assistant *August 2024 to July 2025*

- Wrote a real-time disaster alerting service in Python and PostgreSQL that reads NASA's live event feed, measures the distance from each active wildfire, earthquake, or storm to alumni addresses, and notifies any of 2,000+ alumni sitting in the path, so the office can reach people the same day.
- Alert thresholds live in a settings page the staff change themselves, because the first version needed me for every adjustment and the office could not wait on that. Added batch processing and a dashboard putting active events and affected alumni on a map.
- Set up CI/CD with GitHub Actions deploying to AWS, with environment-scoped credentials and a one-hour cache on the NASA feed, which was what had been driving the external API calls up.
- Built a full-stack inventory system for the same office in React and Node. Fuzzy search and better category tagging cut item lookup time roughly in half.

Nepal Technical Service Butwal, Nepal

Full Stack Engineer *January 2023 to August 2025*

- Built the retail and accounting platform the business runs on, in React, Next.js, TypeScript, Node, and PostgreSQL, with REST APIs and card payments wired in. It has carried 200+ customers and over $40K in transactions since launch, and it replaced the spreadsheets the shop floor ran on.
- Shipped it as a multi-tenant system, one installation serving several stores at once, across six modules for sales, inventory, accounting, ledger, expenses, and payments. Every database query is scoped to a single store, so no shop sees another shop's numbers.
- Every write to the books goes through one guarded path that fails closed. A retried request or a dropped connection cannot leave the accounts half-updated, so the day's totals are still right the next morning.

Projects

BlueHabit, Accessible Route Planner on ESRI GeoAI

- Route planner that finds walkable paths for wheelchair users and residents with vision loss by combining air quality and satellite data on ESRI GeoAI with Claude Sonnet. Fourth place out of 30+ teams at the CGU Ethical AI Hackathon, built over a weekend with three teammates.

GeoAssist, Natural Language Explorer for ArcGIS Feature Services

- Built a React and TypeScript app that lets someone query public ArcGIS Feature Services for hospitals, schools, and cities by typing a plain English question instead of learning REST query syntax. A language model turns the question into a structured request with WHERE clauses, field selection, and sorting, and results render as map graphics with popups.
- Interface is built entirely on Esri's modern web components and the Calcite Design System rather than legacy widgets, with ARIA live regions, keyboard navigation, light and dark theming, and reduced-motion support, because a map nobody can operate with a keyboard is not finished.

Education

Claremont Graduate University Claremont, CA

MS in Information Systems and Technology | AI-Driven Innovation *May 2027*

- Coursework: Software Development, Database Systems, Natural Language Processing, Building Generative AI Apps, Communications and Networking

Soka University of America Aliso Viejo, CA

BA in Economics and Finance *May 2024*

- Coursework: Financial Modeling, Regression Analysis, Business Statistics, Financial Economics